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1. Carboxylation (fixation)
2. Reduction (Glycolytic Reversal)
3. Regeneration

Phase 1- Carboxylation (Fixation)

The acceptor molecule Ribulose 1,5 Bisphosphate (RUBP) a 5 carbon compound with the help of RUBP carboxylase oxygenase (RUBISCO) enzyme accepts one molecule of carbon dioxide to form an unstable 6 carbon compound. This 6C compound is broken down into two molecules of 3-carbon compound phospho glyceric acid (PGA) (Figure 13.19).

$$RUBP + CO_2 \xrightarrow{\text{Rubisco}} 2 \text{ molecules PGA}$$

Phase 2 – Glycolytic Reversal / Reduction

Phospho glyceric acid is phosphorylated by ATP and produces 1,3 bis phospho glyceric acid by PGA kinase. 1,3 bis phospho glyceric acid is reduced to glyceraldehyde 3 Phosphate (G-3-P) by using the reducing power NADPH + H^+. Glyceraldehyde 3 phosphate is converted into its isomeric form di hydroxy acetone phosphate (DHAP).

$$PGA \xrightarrow[ATP \quad ADP]{\text{PGA Kinase}} 1,3 \text{ bisphosphoglyceric acid}$$

$$1,3 \text{ bisphosphoglceric acid} \xrightarrow[NADPH + H^+ \quad NADP^+]{} \text{Glyceraldehyde-3-Phosphate}$$

Phase 3 – Regeneration

Regeneration of RUBP involves the formation of several intermediate compounds of 6-carbon, 5-carbon,4-carbon and 7- carbon skeleton. Fixation of one carbon dioxide requires 3 ATPs and 2 NADPH + H^+, and for the fixation of 6 CO_2 requires 18 ATPs and 12 NADPH + H^+ during C_3 cycle. One 6 carbon compound is the net gain to form hexose sugar.

$$RU5P \xrightarrow[ATP \quad ADP]{} RUBP$$

Overall equation for dark reaction:

$$6CO_2 + 18ATP + 12NADPH + H^+ \longrightarrow C_6H_{12}O_6 + 6H_2O + 18ADP + 18Pi + 12NADP^+$$

RUBISCO – RUBP Carboxylase Oxygenase enzyme, is the most abundant protein found on earth. It constitutes 16 % of the chloroplast protein. It acts as carboxylase in the presence of CO_2 and oxygenase in the absence of CO_2.

13.14 Hatch & Slack Pathway or C_4 Cycle or Dicarboxylic Acid Pathway or Dicarboxylation Pathway

Till 1965, Calvin cycle is the only pathway for CO_2 fixation. But in 1965, **Kortschak**, **Hart** and **Burr** made observations in sugarcane and found C_4 or dicarboxylic acid pathway. Malate and aspartate are the major labelled products. This observation was confirmed by **Hatch** & **Slack** in 1967. This alternate pathway for the fixation of CO_2 was found in several tropical and sub-tropical grasses and some dicots. C_4 cycle is discovered in more than 1000 species. Among them 300 species belong to dicots and rest of them are monocots. C_4 plants represent about 5% of